

Logan Kojiro

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WORK EXPERIENCE

Ampere Computing — Verification Engineer

Santa Clara, CA | Aug 2021 – Mar 2026

- Co-built and served as secondary owner of an internal Python coherency-checking tool modeling Ampere's mesh network — adopted by 90%+ of the mesh DV team and run alongside thousands of weekly regression tests, catching hundreds of bugs in the team's proprietary hierarchical coherency model.
- Debugged test failures, working with designers to fix RTL bugs or working with the team to fix bugs in our checkers
- Architected a C++ DPI layer enabling mid-run bug detection in simulation and emulation, letting engineers kill failing runs early, cutting wasted compute on long overnight regressions.
- Contributed to a company-wide quarterly "Kilowatt" award for our team of 3, recognizing the tool's impact on debugging turnaround time.

ManTech — CNO Developer Intern

Herndon, VA | May 2019 - Aug 2019, May 2020 - Aug 2020

- Built a JavaScript browser-fingerprinting tool that identified and enumerated features of unknown browsers with over 90% accuracy, plus a companion suite of Python tools to verify and extend its functionality.
- Wrote position-independent code in Assembly and C to extract target operating system information from within a browser sandbox.

ARIN Technologies — Software Engineer Intern

Pittsburgh, PA | May 2018 - Aug 2018, Feb 2019 - May 2019

- Designed and implemented a scalable, user-friendly GUI that cut calibration time for ARIN's indoor positioning system by 95% compared to manual calibration.

CMU BioRobotics Lab — Research Assistant

Pittsburgh, PA | Feb 2017 - Aug 2017

- Collaborated with an interdisciplinary team of engineers and biologists to emulate and analyze snake locomotion and behavior in a robotic platform.

EDUCATION

Carnegie Mellon University

Pittsburgh, PA | May 2021

B.S. Electrical & Computer Engineering

PROJECTS

Terminal Tracer — C++20 Real-Time 3D Rasterizer

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- Software rasterizer rendering real-time graphics from .obj mesh files as ASCII artwork in the terminal
- Implemented Z-buffering, MSAA anti-aliasing, and Lambertian shading for accurate depth handling and surface lighting.
- Plays back videos converted to ascii

GPU Fluid Simulation — SPH Solver in CUDA & OpenMP

Parallel Computing and Architecture Final Project

- Implemented a Smoothed Particle Hydrodynamics fluid solver in three variants — sequential, OpenMP, and CUDA
- Designed a Z-order spatial grid with a GPU-optimized compact variant for optimized neighbor searches
- Documented speedups in a real-time simulation of 100,000+ particles across two machines, observing over 200,000x CUDA speedup compared to our sequential implementation.

TECHNICAL SKILLS

Languages: Python, C, C++, CUDA, Assembly

Real Time & Interactive Systems: Real-time rasterization pipeline, Path Tracing (Monte Carlo Integration), GPU-accelerated rendering, 3D geometry & mesh processing, Inverse Kinematics (Jacobian Inverse)

Systems & Tools: Unix/Linux, Git, OpenMP, Protobuf, CHI & AXI Protocol, Parallel Computing, Cache Coherency, ARM Architecture

LEADERSHIP & ACTIVITIES

Carnegie Mellon ECE Department — Teaching Assistant | Sigma Phi Epsilon Fraternity — Greek Sing Chair, House Manager | Boy Scouts of America — Eagle Scout, Patrol Leader, Instructor